WORKED SOLUTION

STEP II 2010, Question 8

Pure • Areas Between Curves

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q8

The key idea

Since $\sin x \leq 1$, the curve $y = \mathrm{e}^{-x}\sin x$ can never rise above $y = \mathrm{e}^{-x}$; it meets it exactly where $\sin x = 1$, and there it just touches. So each area is a single integral of $\mathrm{e}^{-x}(1 - \sin x)$ between consecutive touching points, with no sign changes to worry about. The integral of $\mathrm{e}^{-x}\sin x$ is a standard "integrate by parts twice" job, and because the limits are all points where $\sin x = 1$ and $\cos x = 0$, the antiderivative collapses to something very simple there.

Solution

The sketch

C_1 is the familiar decaying exponential, starting at $(0, 1)$ and decreasing towards 0. C_2 is a damped sine wave: it starts at the origin, is zero at every multiple of π, and is trapped between $y = \mathrm{e}^{-x}$ and $y = -\mathrm{e}^{-x}$ because $-1 \leq \sin x \leq 1$. The oscillations shrink very fast; already the first negative dip has depth only about 0.014 (at $x = \frac{5\pi}{4}$).

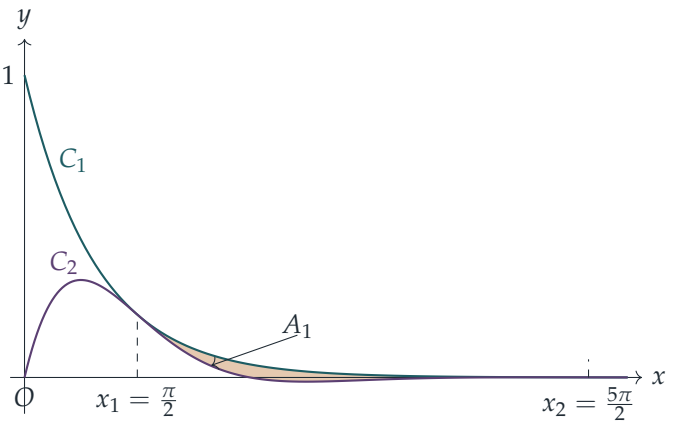
Where do the curves meet? Since $\mathrm{e}^{-x} > 0$,

$$\mathrm{e}^{-x}\sin x = \mathrm{e}^{-x} \iff \sin x = 1 \iff x = \tfrac{\pi}{2} + 2k\pi, \quad k = 0, 1, 2, \ldots$$

At these points the curves touch rather than cross: C_2 lies on or below C_1 everywhere. Indeed the gradients agree there too, since

$$\frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{-x}\sin x) = \mathrm{e}^{-x}(\cos x - \sin x) = -\mathrm{e}^{-x} \quad \text{when } \sin x = 1,\ \cos x = 0.$$

Notice that the first maximum of C_2 is at $x = \frac{\pi}{4}$ (where $\cos x = \sin x$), *before* the first contact at $x = \frac{\pi}{2}$: the sketch should show C_2 peaking and then coming down to meet C_1 tangentially.



The areas A_n

The nth contact point is the nth solution of $\sin x = 1$ with $x > 0$:

$$x_n = \tfrac{\pi}{2} + 2(n-1)\pi = \tfrac{(4n-3)\pi}{2}, \qquad x_{n+1} = x_n + 2\pi.$$

Between x_n and x_{n+1} we have $\mathrm{e}^{-x} - \mathrm{e}^{-x}\sin x = \mathrm{e}^{-x}(1-\sin x) \geq 0$, so C_1 is the upper boundary throughout and

$$A_n = \int_{x_n}^{x_{n+1}} \mathrm{e}^{-x}(1-\sin x)\,\mathrm{d}x.$$

? **Ask yourself** How do we integrate $\mathrm{e}^{-x}\sin x$?

Integrate by parts twice and you get back to the integral you started with. Let $I = \int \mathrm{e}^{-x}\sin x\,\mathrm{d}x$. Integrating by parts with $u = \sin x$, $\mathrm{d}v = \mathrm{e}^{-x}\,\mathrm{d}x$, and then again with $u = \cos x$,

$$\begin{aligned} I &= -\mathrm{e}^{-x}\sin x + \int \mathrm{e}^{-x}\cos x\,\mathrm{d}x \\ &= -\mathrm{e}^{-x}\sin x - \mathrm{e}^{-x}\cos x - \int \mathrm{e}^{-x}\sin x\,\mathrm{d}x, \end{aligned}$$

so $2I = -\mathrm{e}^{-x}(\sin x + \cos x)$ and $I = -\frac{1}{2}\mathrm{e}^{-x}(\sin x + \cos x) + c$. (You can confirm by differentiating.)

Hence an antiderivative of $\mathrm{e}^{-x}(1-\sin x)$ is

$$F(x) = -\mathrm{e}^{-x} + \tfrac{1}{2}\mathrm{e}^{-x}(\sin x + \cos x).$$

Now the neat part: at every contact point $\sin x = 1$ and $\cos x = 0$, so there

$$F(x) = -\mathrm{e}^{-x} + \tfrac{1}{2}\mathrm{e}^{-x} = -\tfrac{1}{2}\mathrm{e}^{-x}.$$

Therefore, using $x_{n+1} = x_n + 2\pi$,

$$\begin{aligned} A_n = F(x_{n+1}) - F(x_n) &= \tfrac{1}{2}\mathrm{e}^{-x_n} - \tfrac{1}{2}\mathrm{e}^{-x_n - 2\pi} \\ &= \tfrac{1}{2}\mathrm{e}^{-x_n - 2\pi}(\mathrm{e}^{2\pi} - 1). \end{aligned}$$

Finally $x_n + 2\pi = \frac{(4n-3)\pi}{2} + \frac{4\pi}{2} = \frac{(4n+1)\pi}{2}$, so

$$A_n = \tfrac{1}{2}(\mathrm{e}^{2\pi} - 1)\mathrm{e}^{-(4n+1)\pi/2},$$

as required.

The infinite sum

The formula shows that $A_{n+1} = \mathrm{e}^{-2\pi} A_n$, so the areas form a geometric series with first term $A_1 = \frac{1}{2}(\mathrm{e}^{2\pi} - 1)\mathrm{e}^{-5\pi/2}$ and common ratio $r = \mathrm{e}^{-2\pi}$. Since $0 < r < 1$ the series converges, and

$$\sum_{n=1}^{\infty} A_n = \frac{A_1}{1 - \mathrm{e}^{-2\pi}} = \frac{\frac{1}{2}(\mathrm{e}^{2\pi} - 1)\mathrm{e}^{-5\pi/2}}{\mathrm{e}^{-2\pi}(\mathrm{e}^{2\pi} - 1)} = \tfrac{1}{2}\mathrm{e}^{-\pi/2}.$$

Checks

- The total is the whole area between the curves to the right of x_1, so it should equal $\int_{\pi/2}^{\infty} \mathrm{e}^{-x}(1 - \sin x)\,\mathrm{d}x = \lim_{X\to\infty} F(X) - F(\frac{\pi}{2}) = 0 + \frac{1}{2}\mathrm{e}^{-\pi/2}$. It does, with no series needed.
- Numerically, $A_1 \approx 0.10375$ and $\frac{1}{2}\mathrm{e}^{-\pi/2} \approx 0.10394$: almost all the area is in the first region, as the sketch suggests, because $\mathrm{e}^{-2\pi} \approx 0.0019$. Simpson's rule on $\mathrm{e}^{-x}(1 - \sin x)$ over $[\frac{\pi}{2}, \frac{5\pi}{2}]$ gives the same 0.103746.

Common slips

- Taking the contact points to be where C_2 has its maxima. The maxima are at $x = \frac{\pi}{4} + k\pi$; the curves touch at $x = \frac{\pi}{2} + 2k\pi$.
- Counting every solution of $\sin x = \pm 1$, or of $\sin x = 1$ spaced by π: consecutive contacts are 2π apart.
- A sign error in $\int \mathrm{e}^{-x}\sin x\,\mathrm{d}x$. Differentiate your answer to check it.

Beyond STEP

The curve C_2 is the shape of a damped oscillation, the motion of a spring with friction, and the envelope $\pm\mathrm{e}^{-x}$ governs how fast it dies away. Integrals like $\int \mathrm{e}^{-x}\sin x\,\mathrm{d}x$ are much quicker as the imaginary part of $\int \mathrm{e}^{(-1+\mathrm{i})x}\,\mathrm{d}x$, a trick used constantly with differential equations and Laplace transforms in first-year courses. Boyce and DiPrima, *Elementary Differential Equations*, treats damped oscillations in detail.

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